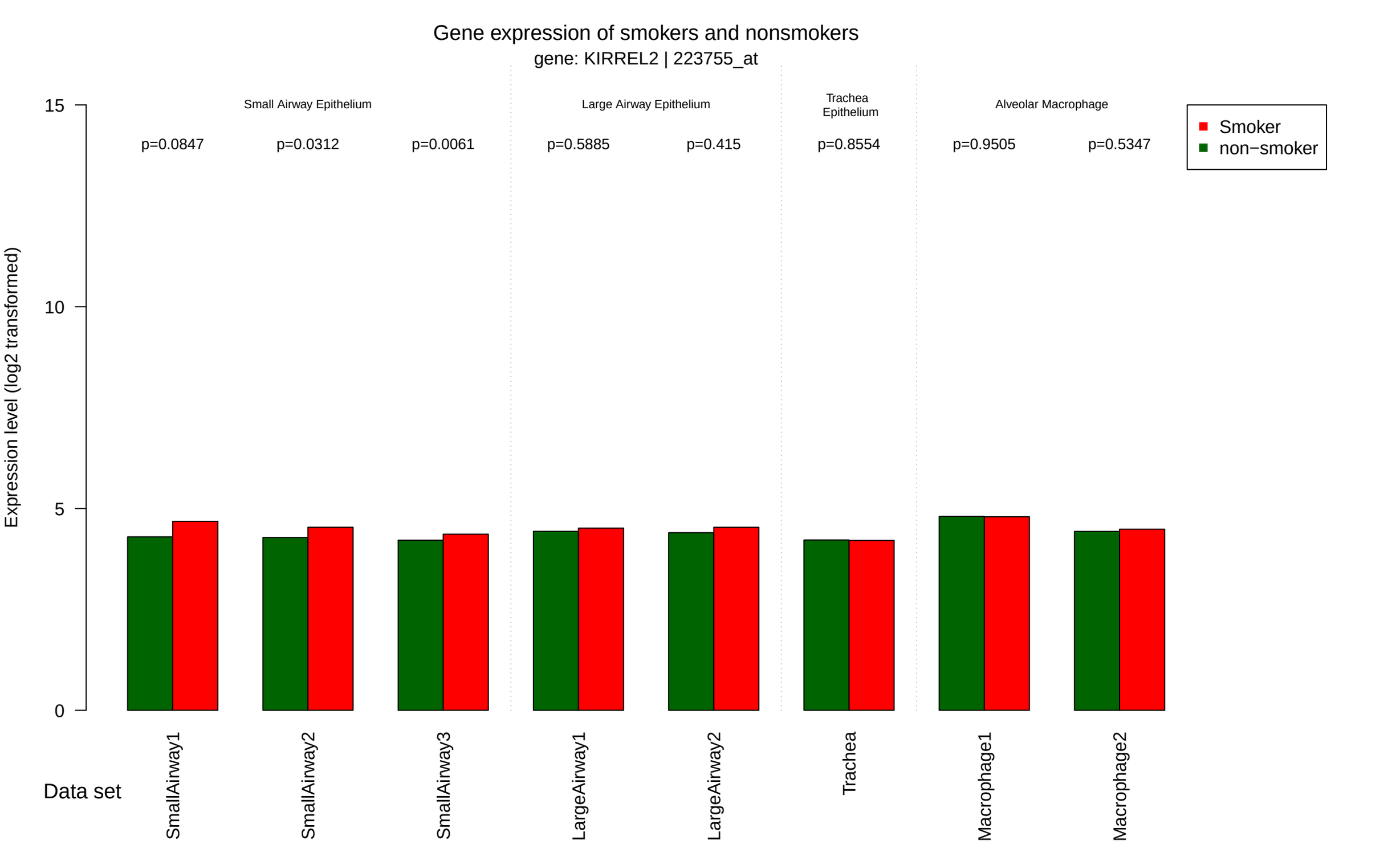
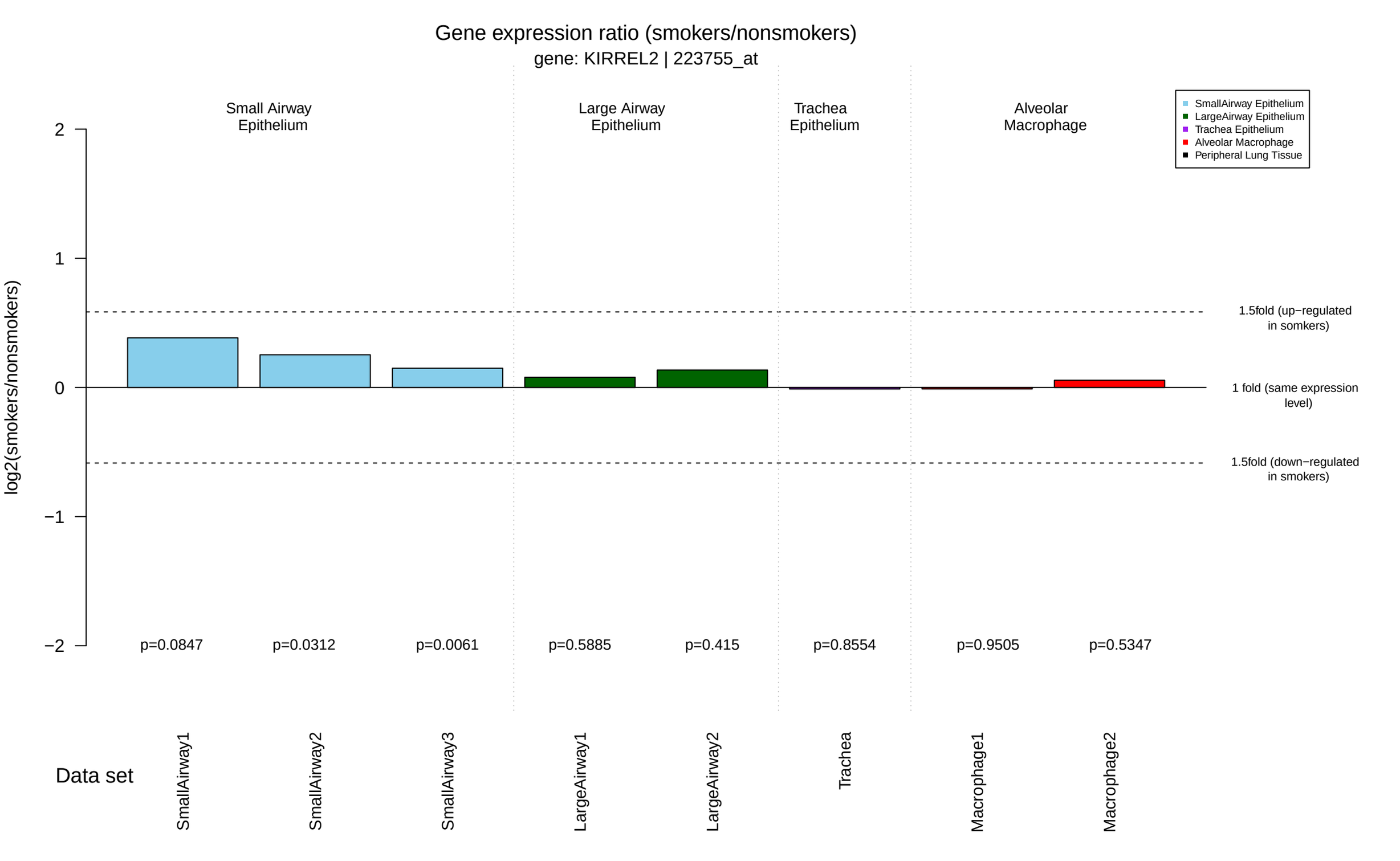
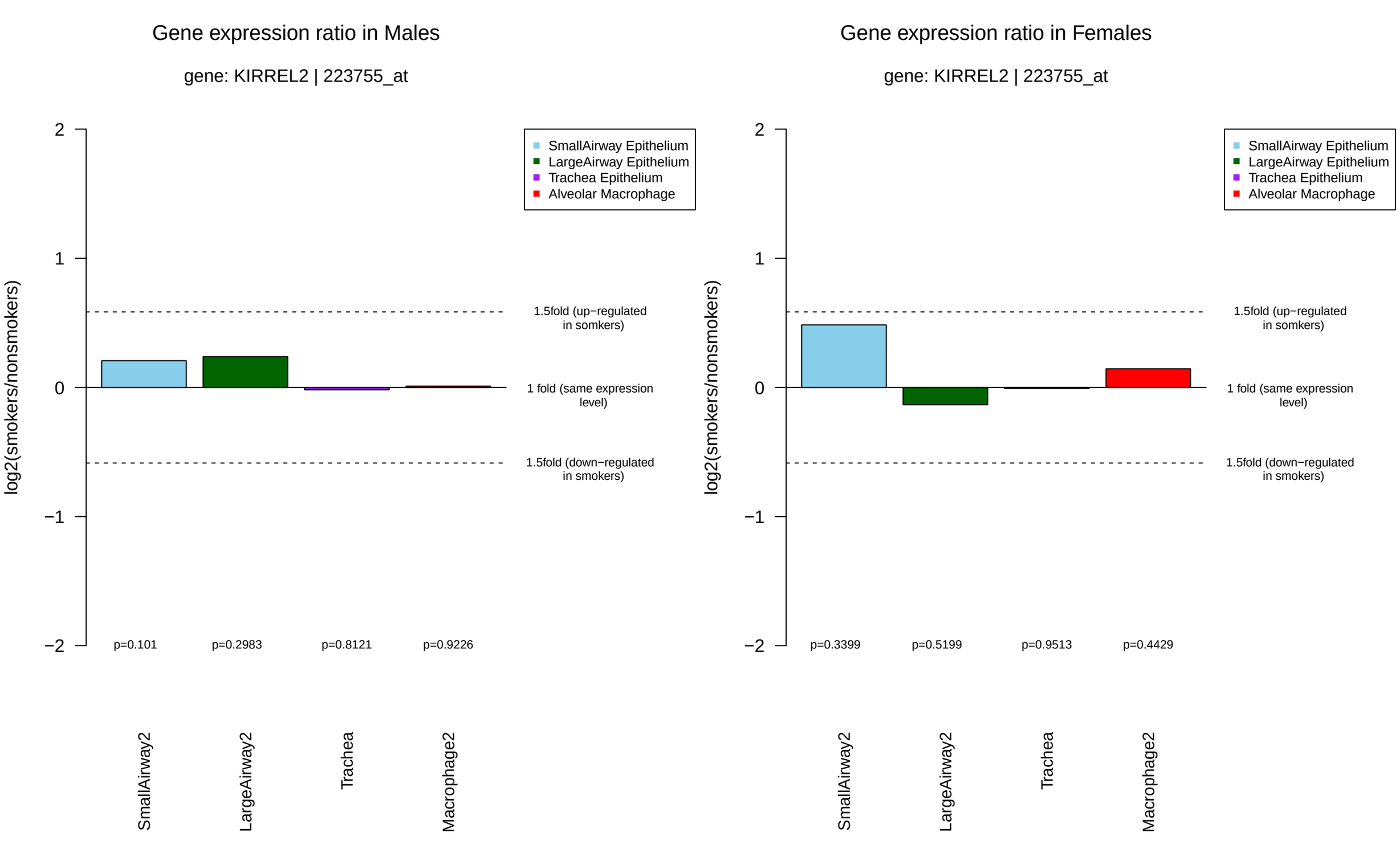
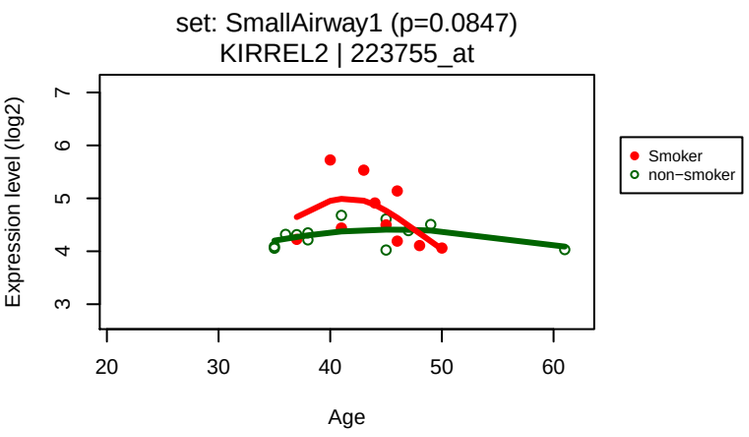
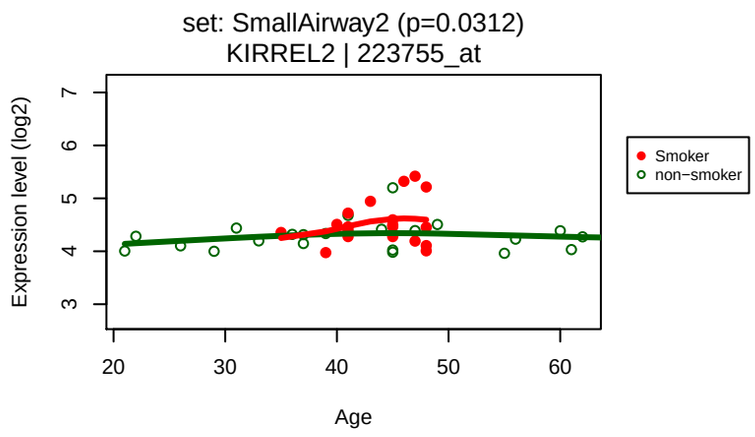
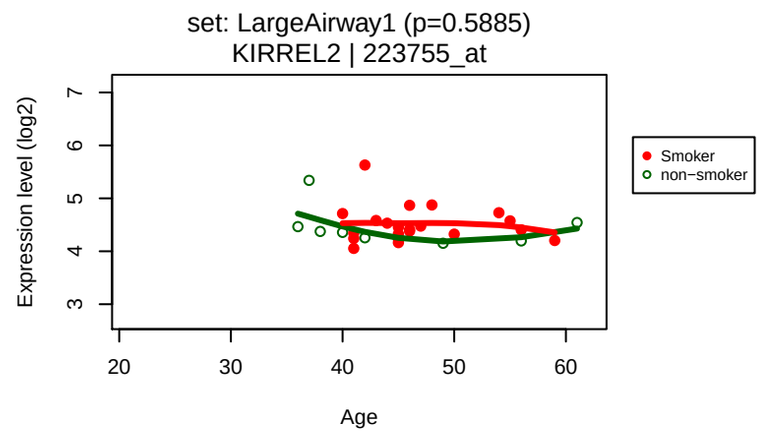

Gene expression of smokers and nonsmokers
gene: KIRREL2 | 223755_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0847
p=0.0312
p=0.0061
p=0.5885
p=0.415
p=0.8554
p=0.9505
p=0.5347
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

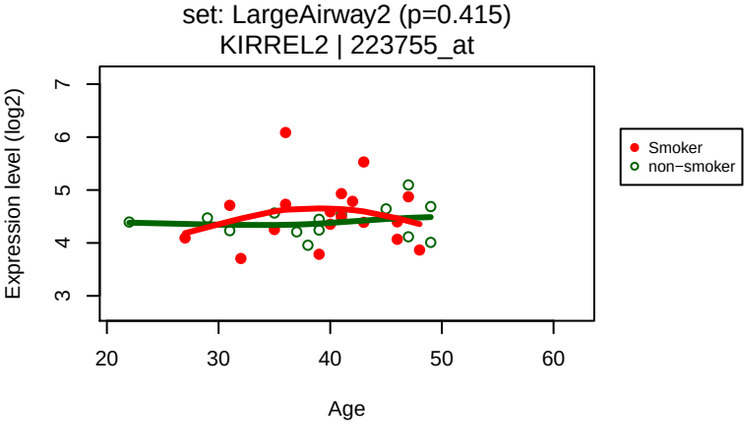

Gene expression ratio (smokers/nonsmokers)
gene: KIRREL2 | 223755_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0847
p=0.0312
p=0.0061
p=0.5885
p=0.415
p=0.8554
p=0.9505
p=0.5347
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

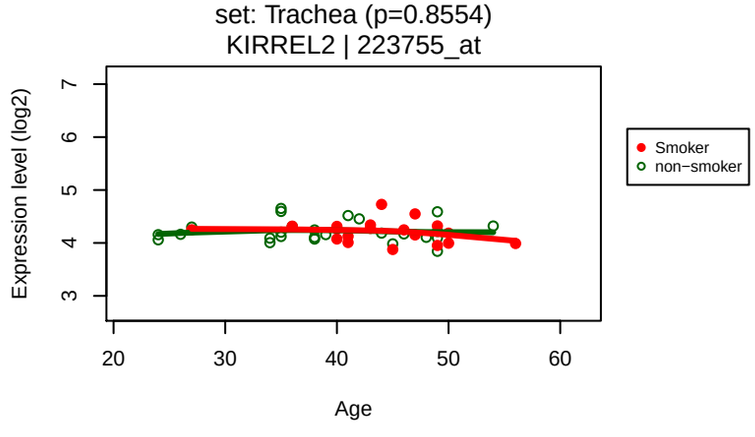

Gene expression ratio in Males
gene: KIRREL2 | 223755_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.101
p=0.2983
p=0.8121
p=0.9226
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIRREL2 | 223755_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3399
p=0.5199
p=0.9513
p=0.4429
SmallAirway2
LargeAirway2
Trachea
Macrophage2

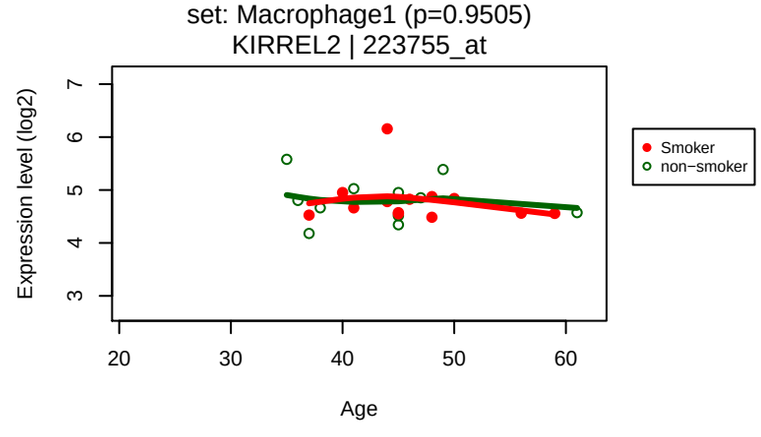

set: SmallAirway1 (p=0.0847)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker

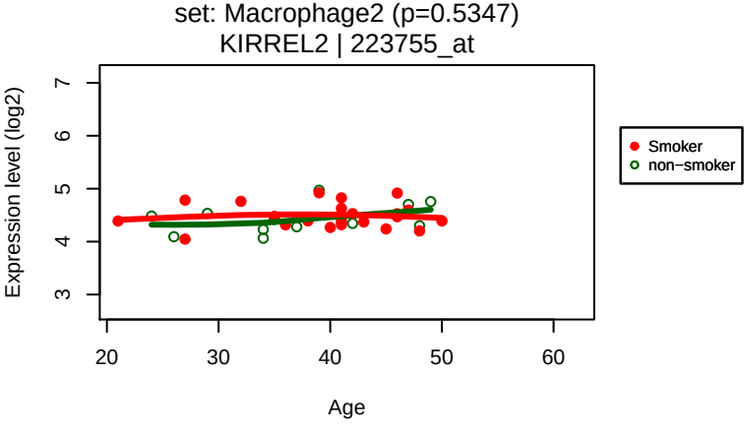

set: SmallAirway2 (p=0.0312)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker

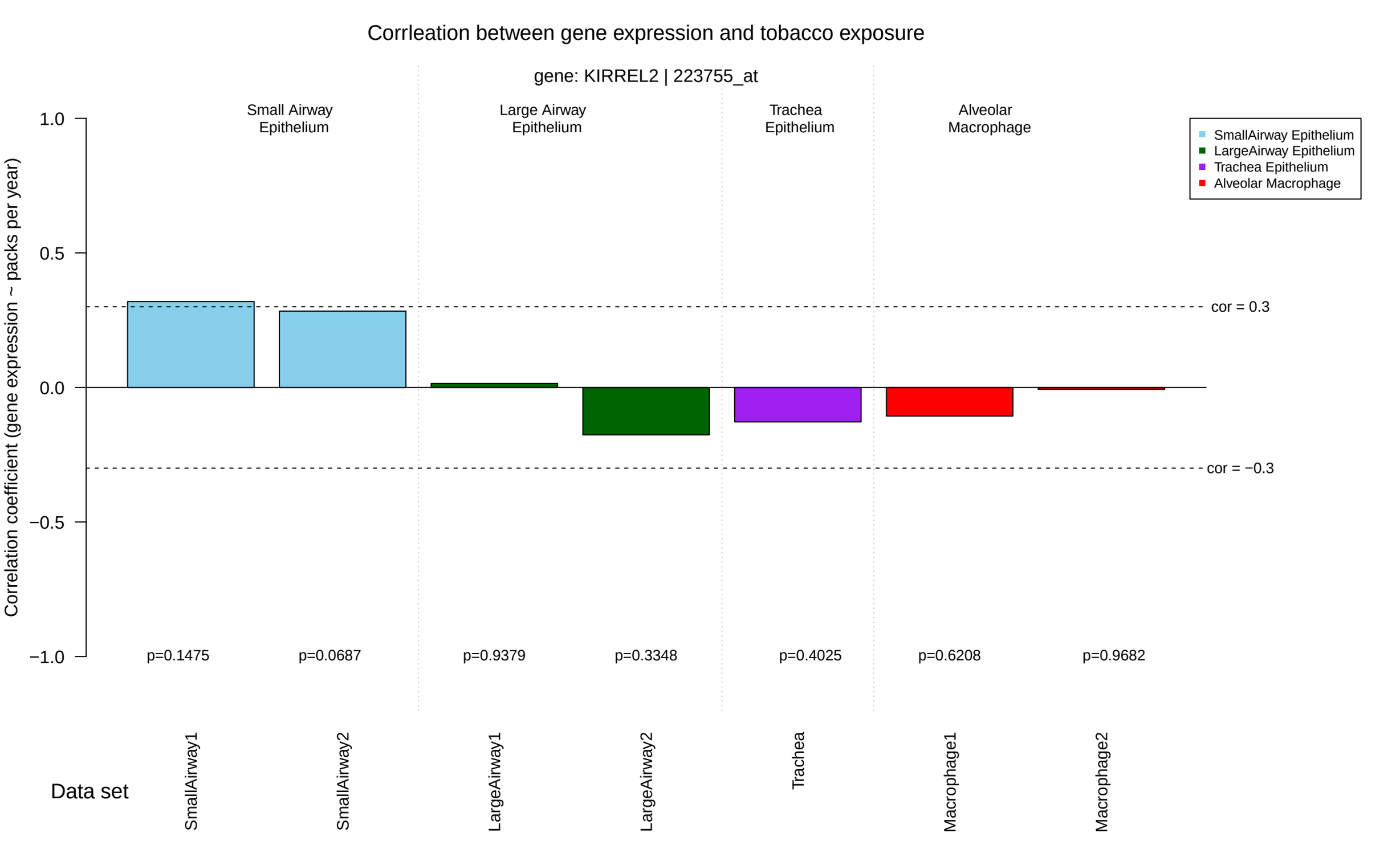

set: LargeAirway1 (p=0.5885)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.415)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8554)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9505)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5347)
KIRREL2 | 223755_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIRREL2 | 223755_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1475
p=0.0687
p=0.9379
p=0.3348
p=0.4025
p=0.6208
p=0.9682
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2